

Basic Sales Data Analysis

Prepared by: Nahla Nabil Dataset: Sample Superstore sales data, 9,994 orders
Tools: Python (Pandas, Matplotlib)

Key Numbers

TOTAL SALES

\$2,297,200.86

TOTAL PROFIT

\$286,397.02

PROFIT MARGIN

12.5%

ORDERS ANALYZED

9,994

1. Data Cleaning

Check	Result	Action Taken
Missing values	Only Postal Code had nulls (11 of 9,994 rows). All analysis columns were 100% complete.	Filled with a placeholder; column isn't used in this analysis.
Duplicate rows	0 fully duplicated rows found.	None needed.
Data types	Date columns were stored as text.	Converted to proper datetime; re-validated all numeric columns (0 failures).
Invalid values	Checked for negative Sales and zero/negative Quantity.	None found, so no rows were removed.

2. Data Analysis: Answers

Sales & Profit by Category

Category	Sales	Profit	Margin
Technology	\$836,154	\$145,455	17.4%
Furniture	\$742,000	\$18,451	2.5%
Office Supplies	\$719,047	\$122,491	17.0%

Sales & Profit by Region

Region	Sales	Profit	Share
West	\$725,458	\$108,418	31.6%
East	\$678,781	\$91,523	29.5%
Central	\$501,240	\$39,706	21.8%
South	\$391,722	\$46,749	17.1%

Top 5 Products by Revenue

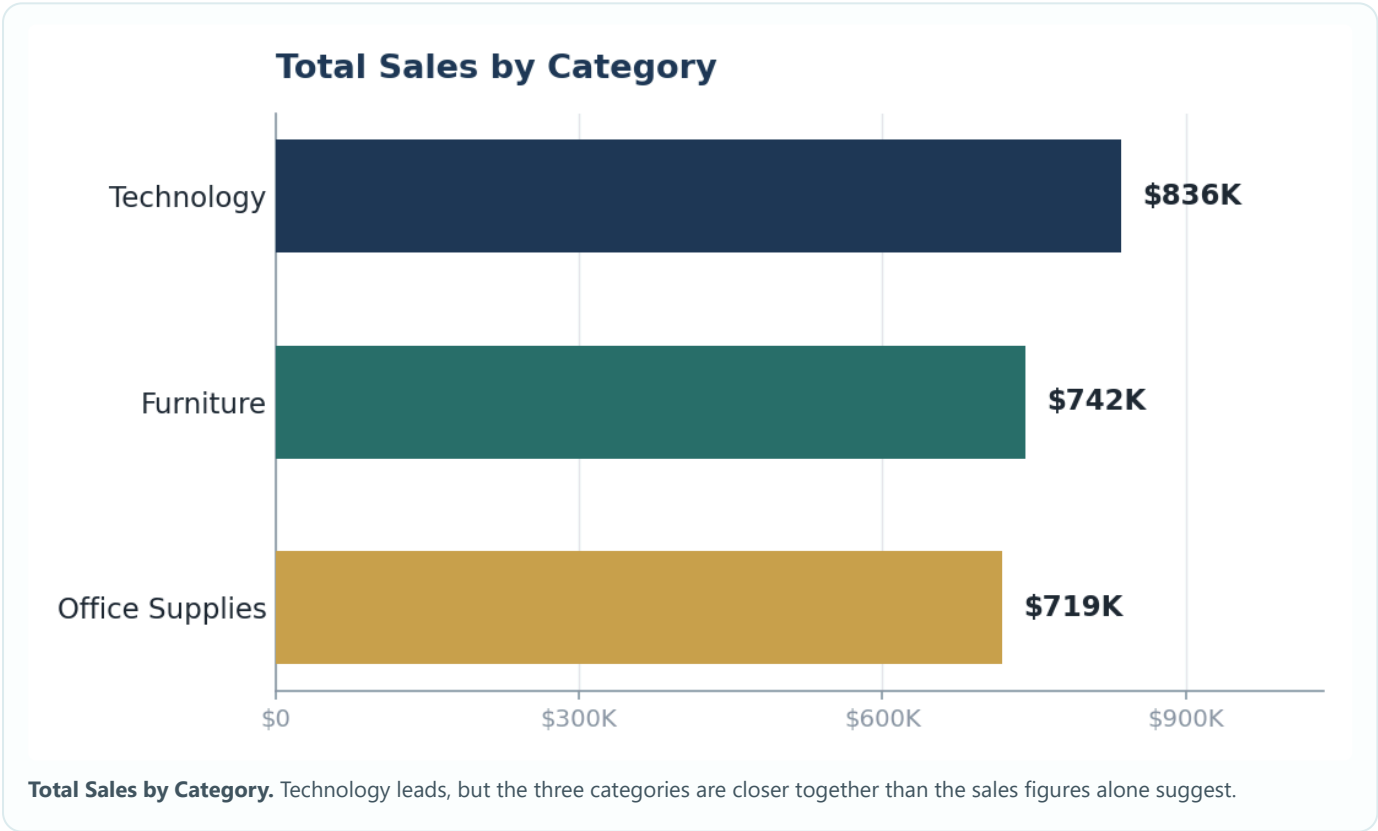
#	Product	Sales
1	Canon imageCLASS 2200 Advanced Copier	\$61,600
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	\$27,453
3	Cisco TelePresence System EX90 Videoconferencing Unit	\$22,638
4	HON 5400 Series Task Chairs for Big and Tall	\$21,871
5	GBC DocuBind TL300 Electric Binding System	\$19,823

Top 5 Products by Units Sold

#	Product	Quantity
1	Staples	215 units
2	Staple envelope	170 units
3	Easy-staple paper	150 units
4	Staples in misc. colors	86 units
5	KI Adjustable-Height Table	74 units

Best sales month: November is the strongest calendar month overall (\$352,461 aggregated across 2015–2018); the single best month on record is 2018-11 (\$118,448).

3. Visualizations

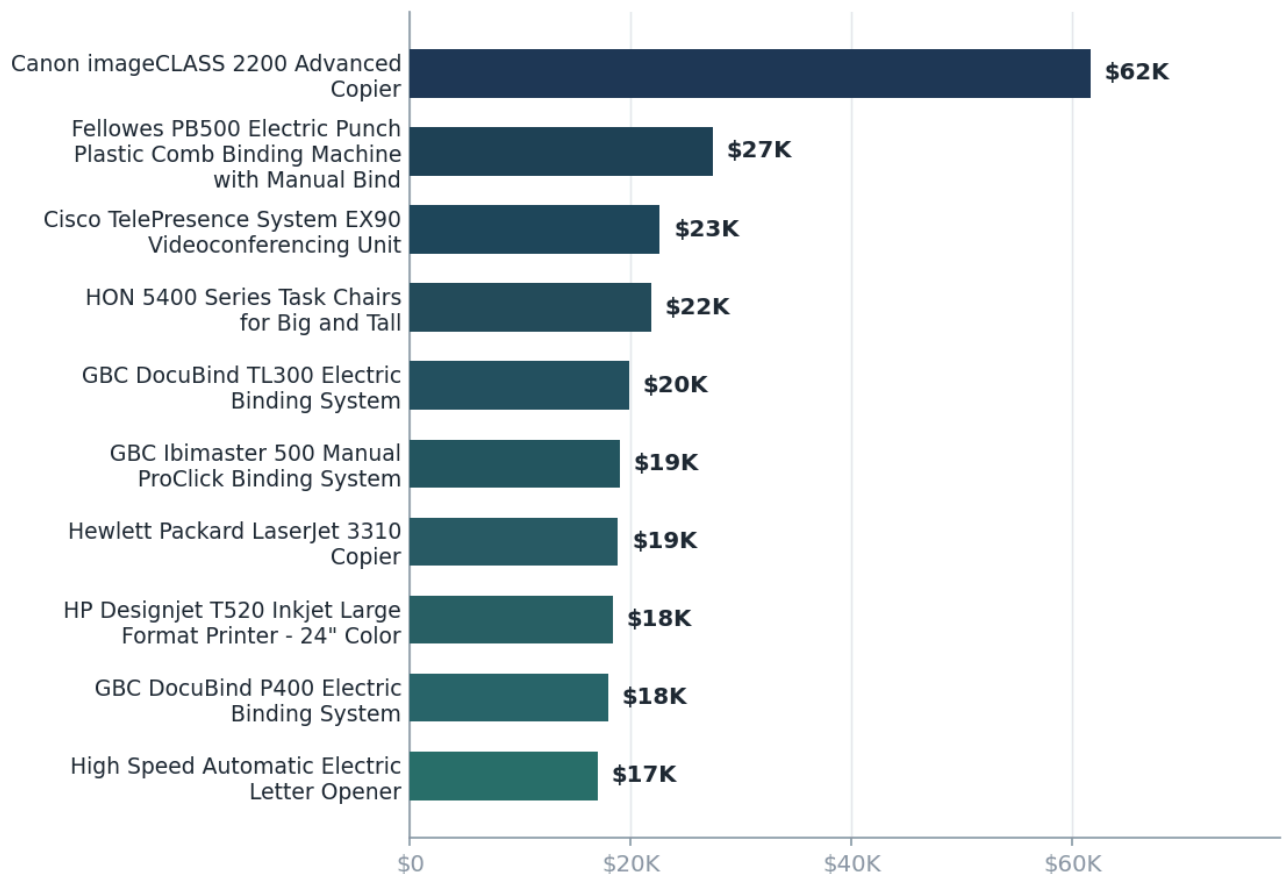


Monthly Sales Over Time (2015-2018)



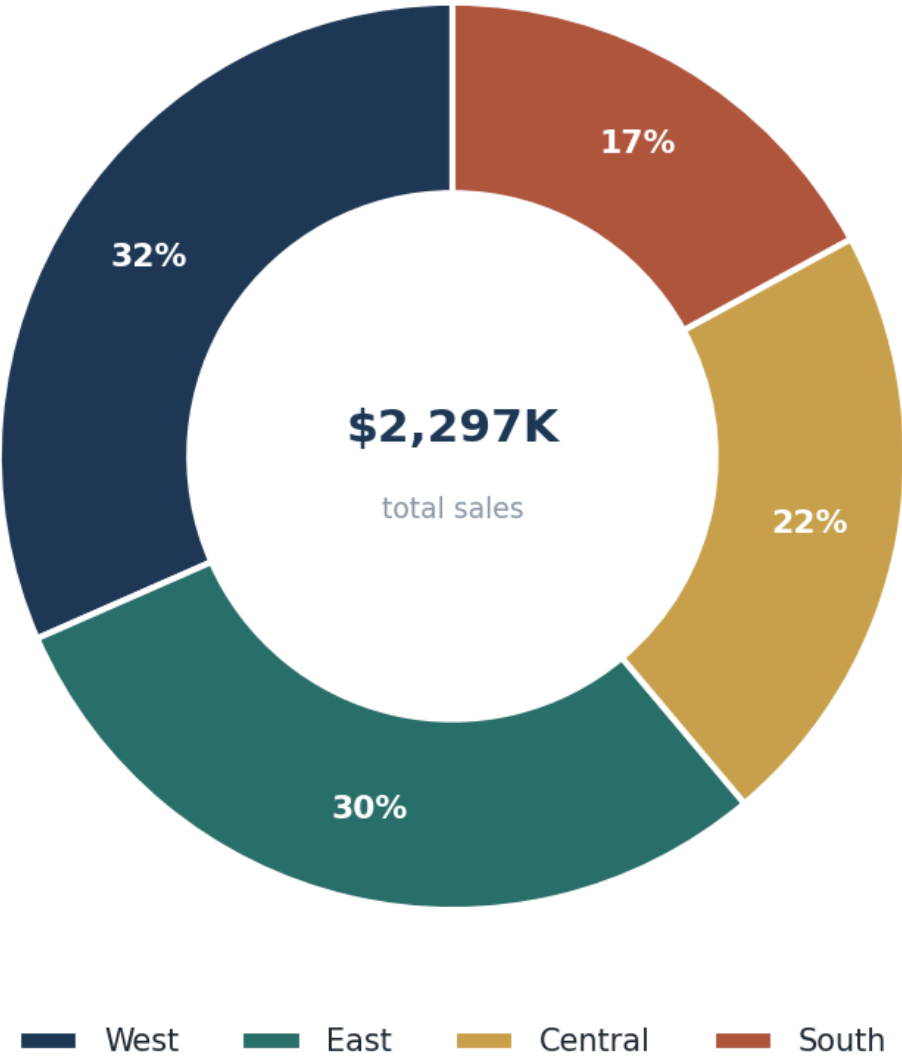
Monthly Sales Over Time (2015-2018). A clear upward trend with recurring year-end spikes, peaking in November 2018.

Top 10 Products by Sales



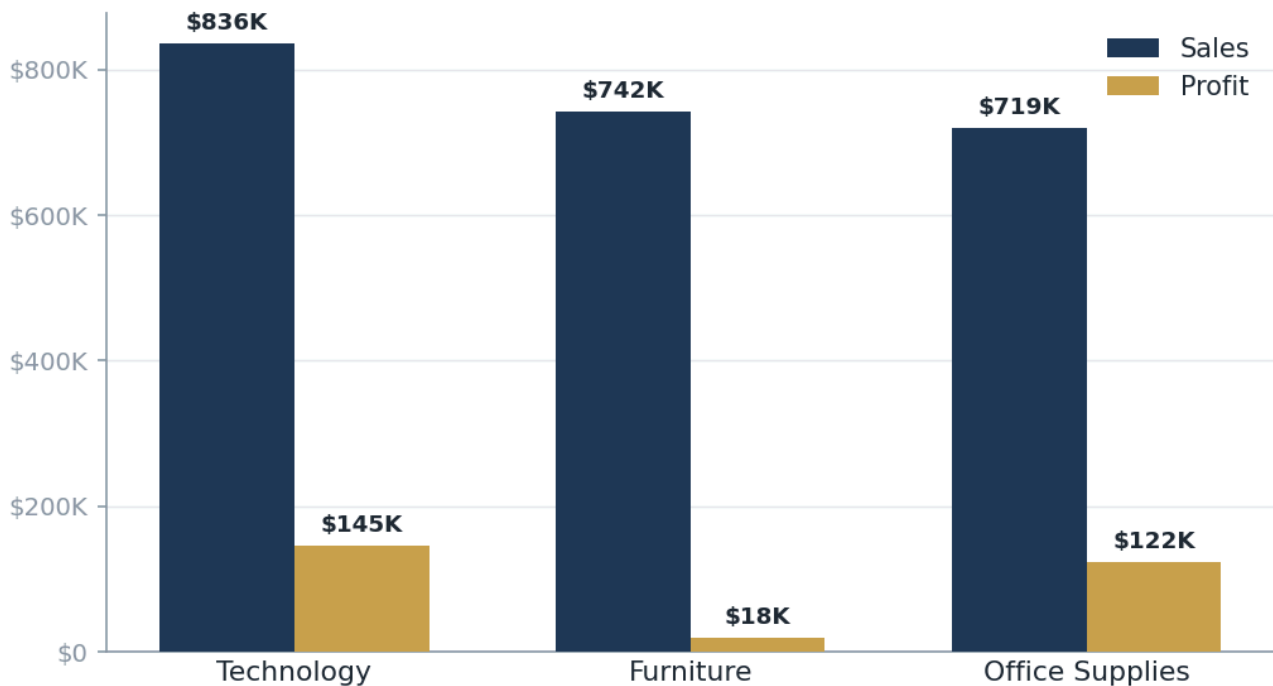
Top 10 Products by Sales. Revenue is concentrated in a handful of high-ticket Technology items, led by the Canon imageCLASS copier.

Share of Sales by Region



Share of Sales by Region. West and East together account for over 60% of all sales, while South lags noticeably behind.

Sales vs Profit by Category



Sales vs Profit by Category. Furniture's profit bar is barely visible next to its sales bar, which is the clearest sign of its weak margin.

4. Insights

1 Technology drives the most revenue, but Office Supplies is just as profitable.

Technology leads sales at \$836,154, but its margin isn't actually any better than Office Supplies. Both categories convert roughly 17.2% of sales into profit, so Technology's lead comes mainly from a handful of high-ticket items like copiers and video-conferencing units, not from stronger performance across the board.

2 Furniture sells well but barely makes money.

Furniture is the #2 category by revenue at \$742,000, but its profit margin is only **2.5%**, compared to roughly 17.2% for the other two categories. That works out to just \$18,451 in profit on those \$742,000 of sales.

3 Tables and Bookcases are actively losing money.

Drilling into sub-categories shows why: **Tables** lost \$17,725 overall and **Bookcases** lost \$3,473, both sold at a net loss despite real revenue behind them. That's the direct cause of Furniture's weak margin above, and a clear place to review pricing or discounting.

4 Sales are heavily seasonal, peaking at year-end.

November is the strongest calendar month overall, with \$352,461 in sales aggregated across all four years, and the single best month on record is **2018-11** at \$118,448. It looks like a holiday-driven demand spike worth planning inventory and promotions around.

5 The West region outsells the South region by nearly 2x.

West and East together generate over 60% of total sales (31.6% and 29.5% respectively), while South trails at just \$391,722, or 17.1% of the total. That gap looks like a real opportunity for targeted growth efforts in the South region.

5. Data Source

This is the widely-used public "Sample Superstore" dataset (9,994 orders; Order Date, Category, Region, Quantity, Sales and Profit among its fields), the same dataset mirrored on Kaggle as <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>.

6. Files in This Submission

- `Sales_Dataset.csv`: the original raw dataset
- `Sales_Dataset_Cleaned.csv`: the cleaned dataset used for analysis
- `analysis.py`: the full Python analysis script
- `analysis_output.txt`: the captured console output and results log
- `Sales_Analysis_Report.pdf`: this report
- `Sales_Analysis_Summary.xlsx`: Excel workbook with pivot tables & charts
- `charts/`: the 5 PNG charts referenced above